

FM-ResXMD: Force-Matched Resolution Exchange Molecular Dynamics Simulation Methods for Enhanced Sampling of Ionic Liquids

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Slow collective relaxation in dense homogeneous liquids is difficult to accelerate with collective-variable-based methods. To address this limitation, we developed force-matched resolution-exchange molecular dynamics (FM-ResXMD), which connects an atomistic model to an on-the-fly coarse-grained interaction learned from mapped atomistic van der Waals forces [1]. Because every replica retains atomistic coordinates, the method eliminates pre-parameterization and backmapping. We applied FM-ResXMD to neat [Bmim][PF₆] and compared it with conventional molecular dynamics and van der Waals HREMD. The learned force tables stabilized within 5 ns. On the same ten-replica ladder spanning $\lambda = 0\text{--}0.9$, FM-ResXMD remained exchange-connected through $\lambda = 0.8$, whereas van der Waals HREMD showed no accepted exchanges and required $\lambda_{\text{max}} = 0.27$ for comparable mean acceptance. FM-ResXMD consequently yielded a low-Q structure-factor peak closer to experiment [2] and greater decorrelation in self and distinct van Hove functions, accompanied by changes in like-ion packing and cation conformations.

To clarify the origin of this enhanced sampling, we first compared fixed- λ force power spectra. Both methods showed a similar redistribution of spectral intensity toward low frequencies as λ increased, indicating that FM-ResXMD does not generate a qualitatively unique force environment at equal λ . We next increased λ_{max} and analyzed the self van Hove functions at the atomistic endpoint. Configurational decorrelation remained limited below approximately 0.5 but increased markedly when λ_{max} reached 0.6–0.7. Finally, to locate this change more quantitatively, we compared relaxation times from fixed- λ van Hove functions with spectral entropy from the force power spectra. The self-overlap relaxation-time curve displayed a distinct bend near $\lambda = 0.65\text{--}0.67$, while spectral entropy showed shallow method-dependent minima at $\lambda = 0.6$ for van der Waals HREMD and 0.8 for FM-ResXMD. Consistently, FM-ResXMD completed three round trips in 20 ns, whereas a 34-replica van der Waals ladder completed none in 120 ns. These results indicate that FM-ResXMD enhances sampling by reaching relaxation-active Hamiltonians while maintaining exchange connectivity across the resolution ladder.

References

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